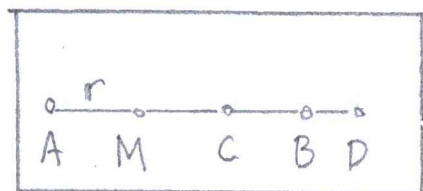


Chapter 7: Independence of the Parallel Postulate:

H-exercises:

H-1.



"case 1"

(from H-exercise) "C and D are harmonic conjugates with respect to AB if and only if we have $r^2 = (MD)/(MC)$ "

Proof: Hypothesis

Evidence

1) Points A and B form a segment

(pg 14) "Definition-segment"

2) Point M is the midpoint on AB

Proposition 4.3

3) Two points C and D are on the same side to M

(pg) "Definition-same side"

4) $r^2 = (MC)(MD)$

"RAA hypothesis"

Case 1: C & D are on the same side to B, but B is between C & D
so $A * M * C * B * D$

$A)(\overline{AB}, \overline{CD}) = \frac{AC}{AD} \cdot \frac{BC}{BD}$

(pg) "Poincare length"

Where $\overline{AC} = 1 + \frac{1}{\sqrt{2}}$
 $\overline{AD} = 1 + \sqrt{2}$
 $\overline{BD} = \sqrt{2} - 1$
 $\overline{BC} = 1 - \frac{1}{\sqrt{2}}$

B) C and D are harmonic conjugates with respect to $\overline{CD}$ because A, C, B, D are collinear and $(\overline{AB}, \overline{CD}) = 1$

Case 2: C & D are on the same side to A and D * A * C * M * B.

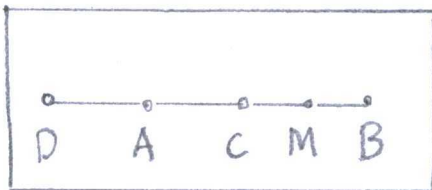
A) $(\overline{AB}, \overline{CD}) = \frac{\overline{AC}}{\overline{AD}} \cdot \frac{\overline{BC}}{\overline{BD}}$

Where $\overline{AC} = 1 - \frac{1}{\sqrt{2}}$

$\overline{AD} = \sqrt{2} - 1$

$\overline{BD} = 1 + \sqrt{2}$

$\overline{BC} = 1 + \frac{1}{\sqrt{2}}$



"Case 2"

B) C and D are harmonic conjugates with respect to $\overline{CD}$ because D, A,

(pg 339) "Definition-
Harmonic Conjugates"

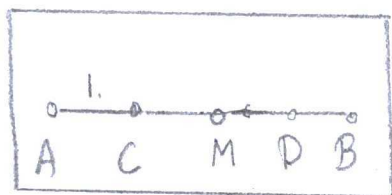
(pg 320) "Poincare
Length"

(pg 339) "Definition-
Harmonic
Conjugate"

C, B are collinear and

$$(\overline{AB}, \overline{CD}) = 1.$$

Case 3: C & D are not
on the same side of M
and $A * C * M * D * B$.



"Case 3"

$$A) (\overline{AB}, \overline{CD}) = \frac{\overline{AC} \cdot \overline{BC}}{\overline{AD} \cdot \overline{BD}}$$

Where $\overline{AC} = 1$

$$\overline{AD} = 2r + 2r^3 + r^2$$

$$\overline{BD} = \frac{1}{2}r^2$$

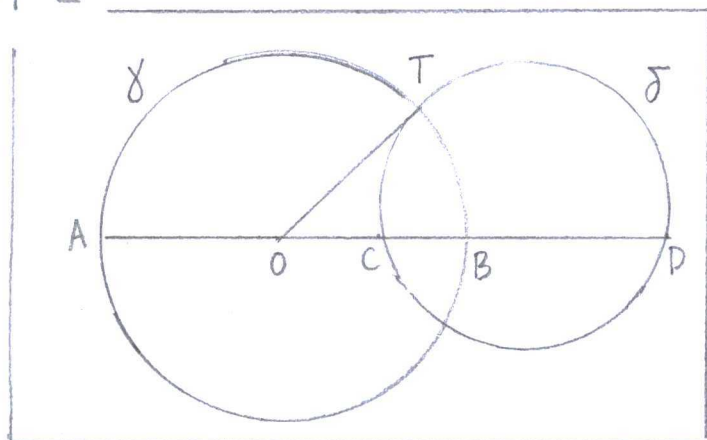
$$\overline{BC} = \frac{1}{2}r^2 + 1$$

B) C and D are not
 harmonic conjugates
 $(\overline{AB}, \overline{CD}) \neq 1$

(pg³²⁰) "Poincare
 Length"

RAA result
 and
 Null hypothesis

H-Z



"Circles are orthogonal"

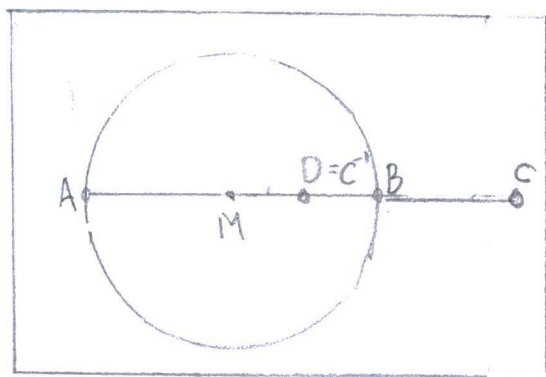
(From H-exercise) "If γ and δ
 are orthogonal circles, if AB
 is a diameter of γ , and if
 δ cuts AB in points C and D ,
 then C and D are harmonic
 conjugates with respect to"

AB: conversely, if a diameter of one circle is cut harmonically by a second circle when the diameter is extended, then the two circles are orthogonal.

Proof: Hypothesis	Evidence
1) From the information given	RAA conjecture
2) T is a point at the intersection between circles γ and δ	Proposition 2.1
3) $(OT)^2 = (OC)(OD)$	Lemma 7.1
4) Point B has an inverse point at A because $(OT)^2 = (OC)(OD)$	(Pg 314) "Definition-Inverse"
5) γ is orthogonal to δ	RAA result and Proposition 7.4

H-3

(From H-exercise) "The fourth harmonic point D is the inverse of C in the circle having AB as the diameter."



"Fourth harmonic point"

Proof: Hypothesis

Evidence

1) Points A and B form a segment

(pg 14) "Definition-Segment."

2) AB has a midpoint M

Proposition 4.3

3) $AM = MB$ is the radius to a circle with AB as diameter

Euclid Postulate III

4) C and D are on the same side to B, where $D \neq B \neq C$

Axiom B-4

$$6) (\overline{MB})^2 = (\overline{MD})(\overline{MC})$$

RAA hypothesis

$$7) (\overline{AB}, \overline{CD}) = \frac{\overline{AC}}{\overline{AD}} \cdot \frac{\overline{BD}}{\overline{BC}}$$

(pg) "Poincare Length"

Where

$$\overline{AC} = 1 + \sqrt{2}$$

$$\overline{AD} = 1 + \frac{1}{\sqrt{2}}$$

$$\overline{BC} = \sqrt{2} - 1$$

$$\overline{BD} = 1 - \frac{1}{\sqrt{2}}$$

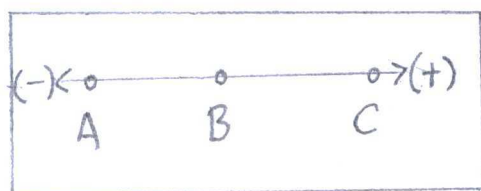
8) C and D are
harmonic conjugates
with respect to
CD because $A \cdot D \cdot B \cdot C$
and $(\overline{AB}, \overline{CD}) = 1$

9) $D \neq M$ and $D = C^{-1}$

(pg 314) "Definition-
Inverse"

RAA result

H-4.



"Sensed
Magnitude"

a) (from H-exercise) "The signed ratio
is independent of the direction
assigned to the line and
that point C is uniquely
determined by this ratio."

Proof: Hypothesis

Evidence

1) A line $\overleftrightarrow{AB}$ forms
from two points;

Euclid Postulate I

$$4) \frac{AC}{CB} = \frac{-CA}{-BC}$$

$$= \frac{AC}{CB}$$

RAA conclusion

b) (from H-exercise) " $\frac{AB}{BC} = \frac{AB'}{B'C'}$ "

A and B.

2) Negative distance
to left of A or
positive distance
to right of B
has no effect on
a ratio $\overline{AC}/\overline{CB}$
by lengths $\underline{AB} = -\underline{BA}$.
("signed length")

Case 1: $A * B * C$

A) $\overline{AC}$ is a segment

B) $\overline{CB}$ is a segment

C) $\overline{AC}/\overline{CB}$

Case 2: $C * A * B$

A) $-\overline{CA}$ is a segment

B) $-\overline{BC}$ is a segment

C) $-\overline{CA}/\overline{BC}$

RAA conjecture

(pg 14) "Definition-
segment"

(pg 14) "Definition-
segment"

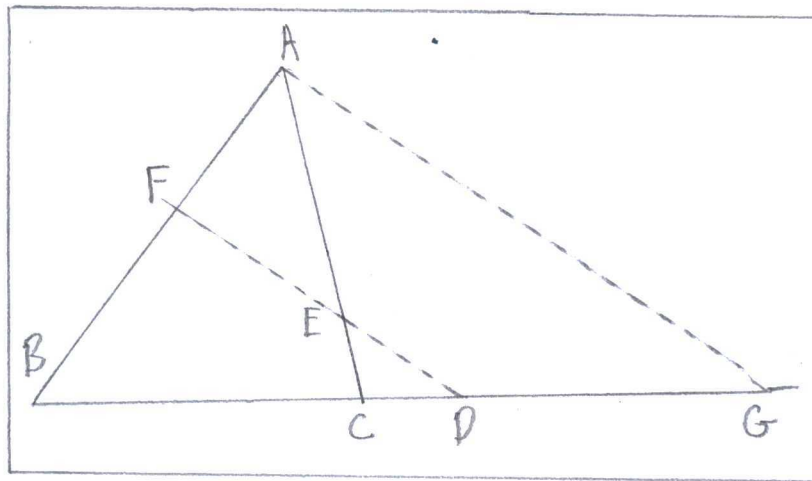
Measurement Theorem

(pg 14) "Definition-
segment"

(pg 14) "Definition-
segment"

Measurement Theorem

H-5



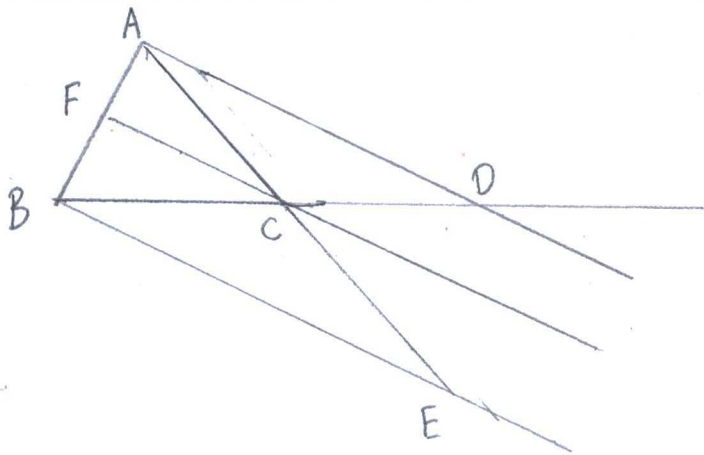
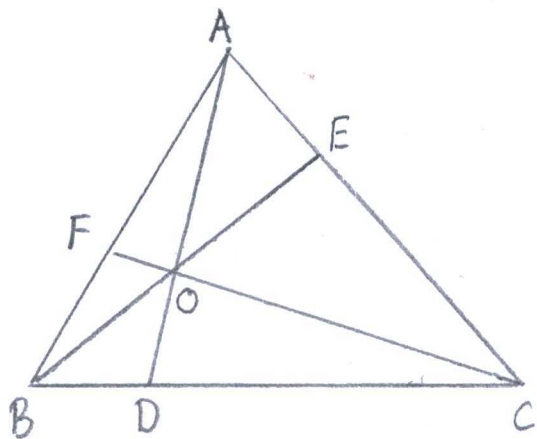
"Theorem of Menelaus"

(from H-exercise) "Given $\triangle ABC$ and points D and G on BC, E on CA, and F on AB that do not coincide with any of the vertices on the triangle"

Note: Linearity Number

$$[ABC/DEF] = (\underline{AF}/\underline{FB})(\underline{BD}/\underline{DC})(\underline{CE}/\underline{EA})$$

H-6



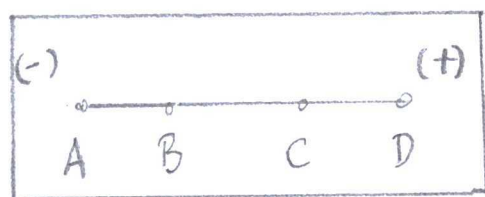
"Theorem of Ceva"

(From H-exercise) "Given $\triangle ABC$ and a third point D on BC . Then the three lines AD , BE , and CF are either concurrent or parallel if and only if $[ABC/DEF] = +1$."

H-7.

(from H-exercise) "signed cross-ratio"

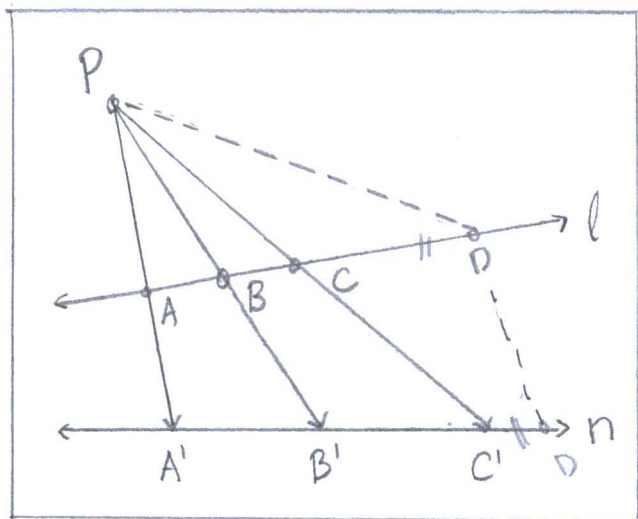
$$(\overline{AB}, \overline{CD}) = -1$$



a) (from H-exercise) "ABCD is a harmonic tetrad if and only if we have $(\overline{AB}, \overline{CD}) = -1$ "

Proof: Hypothesis	Evidence
1) Four collinear points: A, B, C, D	(pg 70) "Definition-Collinear"
2) Positive is right and negative signs are left	RAA hypothesis
3) $(\overline{AB}, \overline{CD}) = \frac{\underline{AC}}{\underline{CB}} \cdot \frac{\underline{DB}}{\underline{AD}}$ $= \frac{\underline{AC}}{\underline{AD}} \cdot \frac{(-\underline{BD})}{\underline{CB}}$ $= -1$	(pg 368) "Definition-Signed cross ratio" RAA result by Harmonic Tetrad (pg 339)

b)



"Perspectivity with point D"

(from H-exercise) "Signed cross-ratios are preserved by perspectivities and parallel postulates"

Proof: Hypothesis
Case 1: Perspectivity

A) Two lines are not parallel; $\vec{l}$ and $\vec{n}$.

B) From point P are four rays across lines $\vec{l}$ and $\vec{n}$

C) The rays intersect $\vec{l}$ at A, B, C, D and $\vec{n}$ at

Evidence

Corollary to Euclid's Postulate V

(pg 18) "Definition-emanate from vertex"

Proposition 2.1

A', B', C', D'

D) Positive sign is the right-direction and negative sign the left-direction

$$E) (AB, CD) = (A'B', C'D')$$

$$F) (\underline{AB}, \underline{CD}) = (\underline{A'B'}, \underline{C'D'})$$

Case 2: Parallel

A) Two lines are parallel; $\vec{n}$ and $\vec{l}$

B) From point P , three rays cross $\vec{l}$ and $\vec{n}$

C) The rays intersect $\vec{l}$ at A, B, C, D and $\vec{n}$ at A', B', C', D'

D) Positive sign is the right-direction and

RAA hypothesis

Lemma 7.5

RAA result

Euclid's Postulate V

(pg 18) "Definition-
Emanate from vertex"

Proposition 2.1

RAA hypothesis

Negative - sign is
in the left-direction.

$$E) (AB, CD) = (A'B', C'D')$$

$$F) (\underline{AB}, \underline{CD}) = (\underline{A'B'}, \underline{C'D'})$$

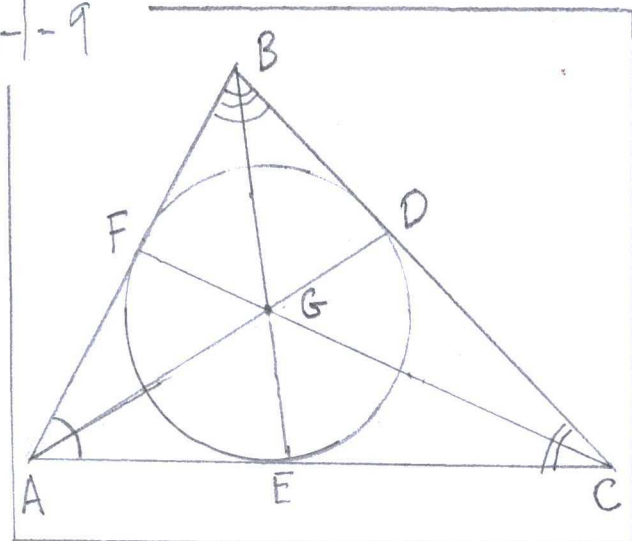
Lemma 7.5

RAA result

H-8

(from H-exercise) "ABCD is a harmonic
tetrad if and only if we have
that $\frac{1}{\underline{AB}} = \frac{1}{2} (\frac{1}{\underline{AC}} + \frac{1}{\underline{AD}})$ "

H-9

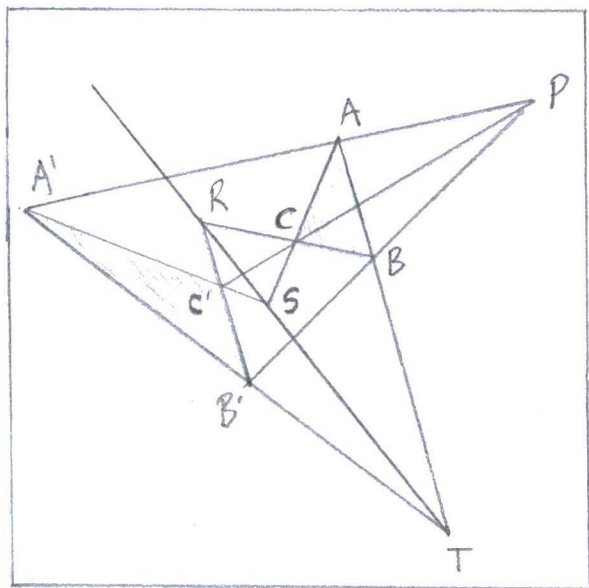


(from H-exercise) "AD, BE, and CF are concurrent in a point G called Gergonne Point of $\triangle ABC$ "

"Gergonne Point"

Proof: Hypothesis	Evidence
1) An inscribed circle of $\triangle ABC$ touches sides AB, BC, and AC at F, D, and E.	Euclid's Postulate IV.4
2) $\angle A, \angle B, \angle C$ have bisectors, α, β, γ	Proposition 4.4(b) (Bisectors)
3) α, β, γ intersect at a concurrent point G	Proposition 2.1
4) AD is α , BE is β and CF is γ	RAA hypothesis

H-10



"Stereographic
Projection"

(pg 93) "Desargues Theorem"

"If the vertices of two triangles correspond in such a way that the lines joining vertices are concurrent, then the intersections of corresponding angles are collinear."

(pg 366) "Theorem of Menelaus"

A necessary and sufficient condition for D, E, and F to be collinear is that $[ABC, DEF] = -1$."

(from H-exercise) "Theorem of Menelaus proves Desargues Theorem"

Proof: Hypothesis	Evidence
1) Desargues depicting	RAA hypothesis

$$\left. \begin{aligned} 2) [BCP/B'C'S] &= \frac{BS}{SC} \frac{CC'}{C'P} \frac{PB'}{B'B} = 1 \\ 3) [CAP/C'A'T] &= \frac{CT}{TA} \frac{AA'}{A'D} \frac{PC'}{C'C} = 1 \\ 4) [ABP/A'B'R] &= \frac{AR}{RB} \frac{BB'}{B'P} \frac{PA'}{A'A} = 1 \end{aligned} \right\} \text{(pg 365) "Definition-Linear Number"}$$

$$\begin{aligned} 5) [BCP/B'C'S][CAP/C'A'T][ABP/A'B'R] \\ &= \frac{BS}{SC} \frac{CT}{TA} \frac{AR}{RB} \\ &= -1 \end{aligned}$$

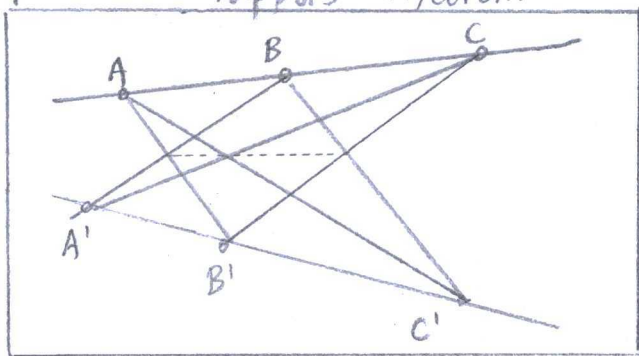
(pg 366) Signed ratio"

$$\begin{aligned} 6) [ABC/RST] &= \frac{AT}{TB} \frac{BR}{DC} \frac{CS}{SA} \\ &= -1 \end{aligned}$$

(pg 339) "Definition-Harmonic Tetrad"

Desargues depicted stereography. The figure had clear alignments between vertices by pencil and ruler.

H-11 "Pappus' Theorem"



(pg 99) "If A, B, and C are three distinct points on one line and if A', B', and C' are three other

distinct points on a second line, then the intersections of lines AC' and CA' , AB' and BA' , and BC' and CB' are collinear."

(from H-exercise) "Use the theorem of Menelaus to prove Pappus' Theorem"

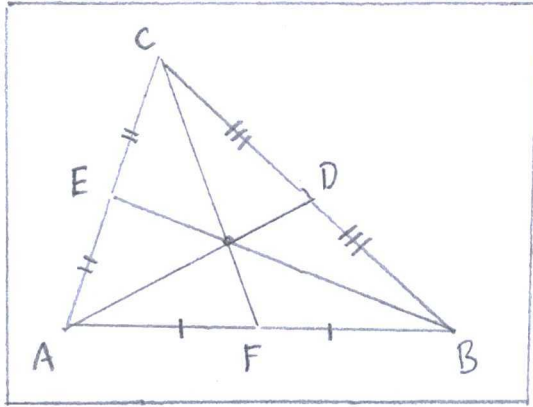
Proof: Hypothesis	Evidence
1) Pappus drawing 2) $N = BA' \cdot BA$ $M = CA' \cdot CA$ $U = BC' \cdot B'C$ $V = CA' \cdot B'A$ $W = CA' \cdot C'B$	RAA hypothesis (pg 364) "Definition-Intersection"
3) $[UW/LCA] = \frac{UA}{AV} \frac{VL}{LW} \frac{WC'}{C'U} = -1$ $[UVW/CMB'] = \frac{VB'}{B'V} \frac{VC}{CW} \frac{WM}{MU} = -1$ $[UVW/A'BN] = \frac{UN}{NV} \frac{VA'}{A'W} \frac{WB}{BU} = -1$ $[UVW/CBA] = \frac{UA}{AV} \frac{VC}{CW} \frac{WB}{BU} = -1$ $[UVW/A'B'C'] = \frac{UC'}{C'V} \frac{VA'}{A'W} \frac{WB'}{B'U} = -1$	(pg 366) "Definition-Linearity Number"

$$4) \left[\frac{UVW}{LCA} \right] \left[\frac{UVW}{CMB} \right] \left[\frac{UVW}{A'BM} \right] \left[\frac{UVW}{CBA} \right] \left[\frac{UVW}{A'BC'} \right]$$

RAA result

$$=-1$$

H-12.



"Medians of
a Triangle"

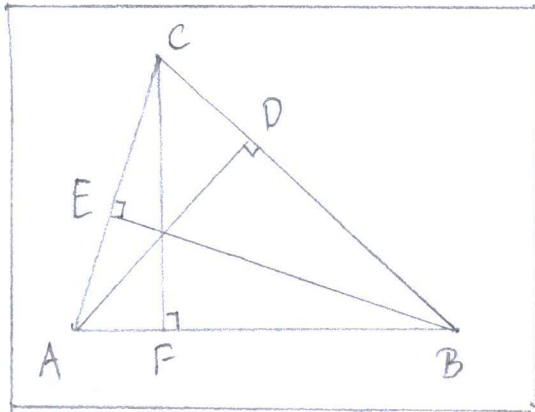
(pg 367) "Ceva's Theorem"

"Given $\triangle ABC$ and a third point D on $\overleftrightarrow{BC}$. Then the three lines AD , BE , and CF are either concurrent or parallel if and only if $[ABC/DEF] = +1$."

a) (from H-exercise) "The medians of a triangle are concurrent."

Proof: Hypothesis	Evidence
1) $\triangle ABC$	(pg 43) "Definition-Triangle"
2) Medians to $\triangle ABC$ by D, E, F	Proposition 4.3 (Midpoints)
3) $[ABC/DEF] = +1$ and so	(pg 366) "Definition-Linearity Number"

b) (from H-exercise) "The altitudes of a triangle are concurrent"



"Altitudes inside a triangle"

Proof Hypothesis

Evidence

1) $\triangle ABC$

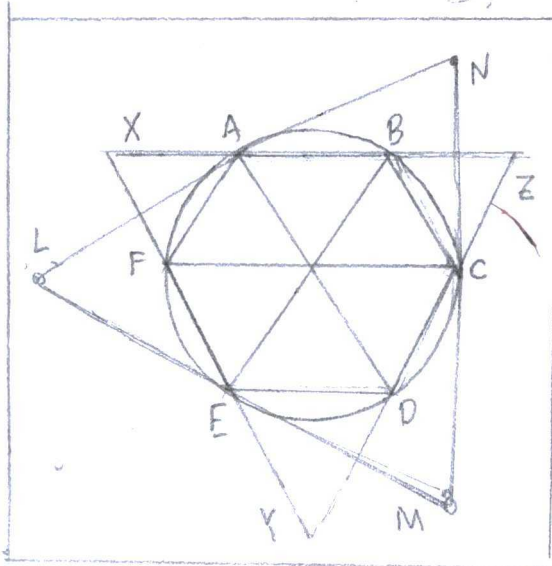
(pg 43) "Definition-Triangle"

2) Altitudes to $\triangle ABC$ by D, E, F

(pg 25) "Definition-Altitudes"

3) $[ABC/DEF] = +1$, thus $\triangle ABC$ has a concurrent point by Ceva's Theorem

(pg 366) "Definition-Linearity Number"



(from H-exercise) "Pascal's Theorem for a hexagon"

"If a hexagon is circumscribed by a circle and the three pairs of opposite sides intersect, then those three points of

intersection are collinear."

Proof: Hypothesis	Evidence
1) A circle with radius $FC/2$	Euclid's Postulate III
2) The circle circumscribes a hexagon at points A, B, C, D, E, and F.	Euclid's Postulate IV.15
3) Points X, Y, and Z are the intersection between AB, CD, and EF.	Proposition 2.1
4) $[XYZ/BCM] = \frac{ZB}{BX} \frac{XC}{CY} \frac{YM}{MZ} = -1$ $[XYZ/AFN] = \frac{ZA}{AX} \frac{XN}{NY} \frac{YF}{FZ} = -1$ $[XYZ/DEL] = \frac{XD}{DY} \frac{YE}{EZ} \frac{ZL}{LX} = -1$	(Pg 366) "Definition- Linearity Number"
5) $[XYZ/BCM][XYZ/AFN][XYZ/DEL]$ $= [XYZ/LNM]$ $= -1$	Multiplication. Crossing